Reversible Symmetric Character Transformation Algorithm

Mathematical Theory and Formulation — Version 2.0

1. Introduction

The proposed algorithm is a reversible, symmetric character transformation method that combines string reversal, position-dependent vector addition, and secret-key integration. Plaintext is transformed into a sequence of bytes through modular arithmetic. Decryption reverses these operations using the same secret key, without requiring additional plaintext information.

2. Mathematical Notations

Symbol	Definition
P	Original plaintext represented as a byte sequence
C	Encrypted ciphertext represented as a byte sequence
S	Secret key
R	Reversed plaintext
A	Intermediate byte sequence
K■	Repeating secret-key byte at position i
n	Plaintext length in bytes
i	Zero-based byte position
M	Modulus (256)

Secret key: Md.Nur-A-Alam

The secret key is repeated cyclically to match the plaintext length. For byte position i, the corresponding key byte is selected using:

```
K_i = S[i mod |S|]
```

All character operations are performed on byte values in the range 0–255. For text containing non-ASCII characters, a consistent byte encoding such as UTF-8 must be used.

3. Encryption Theory

The encryption process consists of three mathematical transformations.

3.1. String Reversal

The plaintext byte sequence is reversed to establish the initial transformation sequence:

```
R_i = P_(n−1−i)
```

3.2. Position-Dependent Vector Transformation

A sequential dependency is introduced between consecutive bytes of the reversed plaintext. The first byte remains unchanged:

```
A_0 = R_0
```

For subsequent positions:

$$A_i = (R_i + R_{(i-1)} + i) \bmod 256, \text{ for } i \geq 1$$

This operation combines the current byte, the preceding byte, and the zero-based position index.

3.3. Secret-Key Integration

The intermediate sequence is combined with the repeating secret key:

$$C_i = (A_i + K_i) \bmod 256$$

The resulting byte sequence represents the encrypted ciphertext.

4. Decryption Theory

Decryption reverses the encryption operations in the opposite order.

4.1. Secret-Key Removal

The secret-key contribution is removed through modular subtraction:

$$A_i = (C_i - K_i) \bmod 256$$

4.2. Reverse Vector Transformation

The reversed plaintext is reconstructed sequentially. The first byte is recovered directly:

$$R_0 = A_0$$

For subsequent positions:

$$R_i = (A_i - R_{(i-1)} - i) \bmod 256, \text{ for } i \geq 1$$

4.3. Original String Restoration

The reconstructed sequence is reversed to obtain the original plaintext:

$$P_i = R_{(n-1-i)}$$

5. Complete Mathematical Formulation

Encryption

$$R_i = P_{(n-1-i)}$$

$$A_0 = R_0$$

$$A_i = (R_i + R_{(i-1)} + i) \bmod 256, \text{ for } i \geq 1$$

$$C_i = (A_i + K_i) \bmod 256$$

Decryption

$$A_i = (C_i - K_i) \bmod 256$$

$$R_0 = A_0$$

$$R_i = (A_i - R_{(i-1)} - i) \bmod 256, \text{ for } i \geq 1$$

$$P_i = R_{(n-1-i)}$$

6. Reversibility Property

The algorithm is designed to reconstruct the original plaintext from its ciphertext using the same secret key. Its mathematical reversibility is expressed as:

$$D_S(E_S(P)) = P$$

Here, E_S represents encryption using secret key S, and D_S represents the corresponding decryption function. The equality holds for valid byte sequences when the ciphertext length and the same secret key are available.

7. Mathematical Characteristics

• **Symmetric key dependency:** The same secret key is used for encryption and decryption.

• **Position dependency:** Each transformed byte depends on its position and the preceding byte in the reversed sequence.

• **Modular arithmetic:** Transformation values remain within the range 0–255.

• **Sequential reversibility:** Each original byte can be reconstructed using the previously recovered byte.

• **Variable-length support:** The repeating secret key accommodates plaintexts of different lengths.

8. Security Qualification

Mathematical reversibility does not imply cryptographic security. This custom transformation has not been independently evaluated and should not be used to protect sensitive information without further cryptographic analysis. In particular, it does not provide authenticated encryption or tamper detection. Established authenticated encryption schemes should be used for production protection of confidential data.